

Course Title	Mathematical Statistics		
Course Code	MC221	Credit Structure	3-1-0-4 (L-T-P-C)
Category	Core	Semester	Winter Semester (AY 25-26)
Program	B.Tech		
Prerequisites	Introductory Probability and Statistics, Calculus		
Course Objectives	This course provides a rigorous introduction to Probability, Sampling Distributions, Estimation, Hypothesis Testing, Regression, and Correlation. The emphasis is on both theoretical foundations and practical applications necessary for undergraduate students in statistics and related fields.		
Course Outcomes	After this course, students will be able to * demonstrate knowledge of, and properties of, statistical models in common use [PO1,PO12] * understand the basic principles underlying statistical inference (estimation and hypothesis testing)[PO2, PO4] * be able to construct tests and estimators, and derive their properties [PO4, PO5] * demonstrate knowledge of applicable large sample theory of estimators and tests. [PO1,PO11] * Understand the assumptions and limitations of non-parametric statistical tests [PO2, PO4]		
Grading Policy	The evaluation components include Assignments, TWO In-Semester examinations, and an End semester exam. All exams are mandatory , and there will be no make-up exams. Missed exams for any reason will result in a grade of zero. The weightage for the evaluation components will be 30% Assignments, 30% In-Semester exams and 40% End Semester.		
Text/Reference Books	1. Statistics for Business and Economics by DR. Anderson et. al, Cengage Learning 2. Introductory Statistics by Sheldon Ross, Elsevier 3. Mathematical Statistics with applications in R by KM Ramachandran, CP Tsokos, Elsevier 4. Mathematical statistics with applications by Dennis D. Wackerly, William Mendenhall, Richard L. Scheaffer, Thomson Brooks/Cole 5. Statistics by David Freedman, Robert Pisani, Roger Purves 6. Statistics for Business and Economics by JT McClave et. Al 7. Introduction to Mathematical Statistics by Robert Hogg, Allen Craig		
Detailed Course Content	See next page...		

Course Outcome:

CO1, CO2, CO3, CO4

POs-COs Matrix:

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
x	x		x	x						x	x

Lecture Schedule

Sl. No.	Description		No. of Lectures
1	Basic Probability		05
	1.1	Review of Probability	
	1.2	Probability distributions	
	1.3	Moment generating functions	
2	Sampling distribution		07
	2.1	Sample, Population and sampling	
	2.2	Probability and the distribution of sample means	
	2.3	Central limit theorem and the law of large numbers	
	2.4	Methods of sampling	
	2.5	Looking ahead to inferential Statistics	
3	Statistical Inference		13
	3.1	Parameter estimation	
	3.2	Types and Criteria for a good estimator	
	3.3	Point estimator	
	3.4	Interval estimator	
	3.5	Testing of Hypothesis	
	3.6	Parametric tests	
4	Non-parametric test		05
	4.1	Introduction	
	4.2	Chi-square tests	
	4.3	Tests for goodness of fit	
	4.4	Tests for independence	
5	Regression analysis		05
	5.1	Introduction	
	5.2	Significance of the Regression equation	
	5.3	Linear and multiple regression	
6	Correlation		05
	6.1	Introduction	
	6.2	Pearson correlation	
	6.3	Hypothesis tests using Pearson correlation	
	6.4	Pearson correlation alternatives	